
CLEAN FEATURE ANCHORING: MITIGATING THE ACCURACY-ROBUSTNESS TRADE-OFF BY SELF-DISTILLING A NATURALLY TRAINED MODEL

Anonymous authors

Paper under double-blind review

ABSTRACT

Adversarial training improves robustness but often sacrifices accuracy on clean data, leading to a persistent clean-robust trade-off. We address this trade-off by first learning a high-accuracy clean self-teacher from natural images and then preserving its clean knowledge while adversarially training the student. We propose Clean Feature Anchoring (CFA), which trains the student to maintain the teacher’s clean representation under adversarial perturbations while retaining the teacher’s classifier. We show that standard knowledge distillation approaches regression onto a fixed clean target as the temperature increases, motivating the direct feature anchoring used in CFA. We further show that keeping adversarial student features close to this clean target simultaneously preserves their agreement with the teacher on clean inputs and limits their variation within the perturbation neighborhood. Our analysis also reveals that higher teacher clean accuracy does not necessarily lead to better clean performance after robust training. Finally, we observe that the same perturbation magnitude can affect the anchoring objective differently across training samples. CFA therefore assigns smaller perturbation radii to more sensitive samples and larger radii to less sensitive ones while keeping the average perturbation budget fixed. Across CIFAR-10, CIFAR-100, and Tiny-ImageNet, CFA achieves state-of-the-art clean-robust trade-offs among the compared methods, retaining high clean accuracy while maintaining strong adversarial robustness.

1 INTRODUCTION

Adversarial training is one of the most effective approaches to improving robustness against adversarial perturbations, but this robustness often comes at the cost of accuracy on clean data (Madry et al., 2017; Tsipras et al., 2018; Zhang et al., 2019). This persistent clean-robust trade-off raises a natural question: can a model acquire adversarial robustness while retaining more of the knowledge that supports strong clean performance?

Several approaches seek to mitigate the clean-robust trade-off by leveraging knowledge from naturally trained models. Such models provide accurate predictions and representations on clean inputs, making them useful references for preserving clean performance during adversarial training. Existing methods incorporate this knowledge through teacher supervision or regularization. However, clean knowledge can still be altered as the student adapts to adversarial perturbations, leaving open how it can be preserved throughout robust training.

We approach this problem by separating the learning of clean knowledge from the subsequent learning of robustness. A clean self-teacher is first trained on natural images and then fixed, providing a stable reference for the student during adversarial training. Importantly, a more accurate clean teacher does not necessarily transfer more clean performance to the robust student: we find that teachers with similar clean accuracy can produce different transfer outcomes, and that increasing teacher accuracy does not consistently improve student clean accuracy. This motivates treating the teacher as a fixed source of clean knowledge rather than assuming that its clean accuracy alone determines the quality of robust transfer.

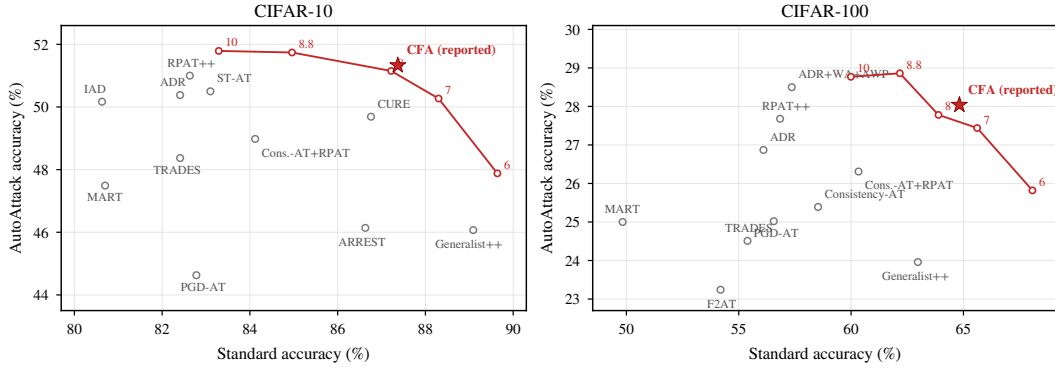


Figure 1: Clean–AutoAttack trade-off on CIFAR-10 and CIFAR-100 with ResNet-18 at $\epsilon = 8/255$. Grey points are published results (Table 27); the CFA curve varies the training radius over 100-epoch runs, and the star denotes the reported 150-epoch result trained at $\epsilon_{\text{train}} = 8/255$.

We instantiate this perspective with **Clean Feature Anchoring (CFA)**. We first train a clean self-teacher on natural images and initialize the student from this model. During adversarial training, CFA matches the student’s representation of an adversarial input to the self-teacher’s fixed representation of the corresponding clean input, while retaining the inherited classifier. The teacher is therefore not required to provide robustness or targets on adversarial inputs. Instead, it serves as a fixed clean reference: the self-teacher defines what clean knowledge should be preserved, while adversarial training determines the neighborhood over which the student learns to preserve it.

CFA anchors the student’s adversarial representations to a fixed clean reference. We show that controlling this anchor simultaneously keeps the student’s clean representation close to the teacher and limits its variation within the adversarial neighborhood.

The same perturbation magnitude, however, can affect the anchoring objective differently across training samples. We therefore introduce *sensitivity-matched perturbation budgets*, assigning smaller perturbation radii to more sensitive samples and larger radii to less sensitive ones while preserving the average perturbation budget.

Across CIFAR-10, CIFAR-100, and Tiny-ImageNet, CFA achieves state-of-the-art clean–robust trade-offs among the compared methods, consistently combining high clean accuracy with strong adversarial robustness.

Our contributions are summarized as follows:

- We introduce a clean self-teacher framework for mitigating the clean–robust trade-off by separating the learning of clean knowledge from the subsequent learning of robustness. CFA instantiates this principle by anchoring adversarial student representations to a fixed clean teacher representation.
- We analyze the role of the fixed clean anchor in adversarial training. We show that controlling the anchor simultaneously keeps the student’s clean representation close to the teacher and limits its variation within the adversarial neighborhood. We further find that teacher clean accuracy alone does not determine how well clean performance is retained after adversarial training.
- We introduce sensitivity-matched perturbation budgets, which redistribute a fixed batch perturbation budget according to the local sensitivity of the anchoring objective, complementing the fixed clean target with example-dependent adversarial neighborhoods.

2 RELATED WORK

Adversarial training and its targets. PGD adversarial training optimizes a model on worst-case inputs (Madry et al., 2017); TRADES separates clean prediction from local consistency (Zhang et al., 2019), and MART reweights misclassified examples (Wang et al., 2020). Subsequent work

improves generalization through label smoothing, consistency regularization, weight averaging, and adversarial weight perturbation (Pang et al., 2020; Tack et al., 2022; Rice et al., 2020; Wu et al., 2020). ADR replaces one-hot labels with targets produced by an annealed momentum teacher (Wu et al., 2024). These approaches motivate our study of how the supervisory target affects the clean-robust trade-off.

Predictive targets in adversarial distillation. ARD transfers a robust teacher’s predictions during adversarial training (Goldblum et al., 2020). RSLAD emphasizes robust soft labels (Zi et al., 2021), IAD discounts unreliable teacher guidance (Zhu et al., 2021), and AdaAD uses the teacher in both adversarial-example generation and student optimization (Huang et al., 2023). IGDM adds the teacher’s local input geometry to logit-based distillation (Lee et al., 2025). These methods are designed to inherit robustness from an adversarially trained teacher. Our target instead comes from a natural teacher at the clean input and remains fixed during the student’s inner maximization.

Guidance from naturally trained models. BGAT is the closest direct-regression precedent: it matches adversarial logits to a fixed natural teacher’s clean logits, and LBGAT extends this to a jointly trained natural branch (Cui et al., 2021). Our logit control confirms that this fixed-clean-target principle is not specific to features. CFA uses the full clean representation with an inherited fixed classifier, and our analysis focuses on what a fixed clean anchor controls, how teacher properties relate to transferability, and how its perturbation neighborhood is allocated. ARREST starts from a naturally pretrained model and regularizes latent representations while retaining adversarial cross-entropy and noisy replay (Suzuki et al., 2023). HAT uses a natural helper to label examples outside the training ball (Rade & Moosavi-Dezfooli, 2022), and B-MTARD combines clean and robust teachers with adaptive temperature and loss balancing (Zhao et al., 2024).

Sample-dependent perturbation budgets. IAAT and CAT adapt the training radius to sample difficulty, while MMA directly increases an input-space margin (Balaji et al., 2019; Cheng et al., 2022; Ding et al., 2020). Our allocation has a different source: it redistributes a fixed batch budget according to the input sensitivity of the loss being optimized. The comparison is therefore not between adaptive and uniform radii alone, but between the signals used to assign the same budget.

3 ANALYSIS

Adversarial training often improves robustness at the cost of clean accuracy. To mitigate this trade-off, we first learn a clean self-teacher from natural images and then use it as a fixed reference while adversarially training the student. A central question is how this clean knowledge can be preserved under adversarial perturbations.

We begin with standard knowledge distillation, which matches temperature-scaled teacher and student output distributions through KL divergence, and examine its high-temperature behavior. We then study the fixed clean target used by CFA and characterize what controlling this anchor implies for the student’s representation. Finally, we examine how teacher clean accuracy relates to the clean performance retained by the adversarially trained student.

3.1 FROM KNOWLEDGE DISTILLATION TO A FIXED CLEAN TARGET

Let $f = h \circ \Phi$, where Φ is a feature map and $h(z) = Wz + b$ is a linear classifier producing logits $f(x) \in \mathbb{R}^C$ for C classes. Subscripts t and s denote the frozen clean self-teacher and student, respectively. For a clean input x and a perturbed input $x' \in B(x, \epsilon)$, standard knowledge distillation with a shared temperature τ matches the teacher and student output distributions

$$q_t^\tau(x) = \text{softmax}(f_t(x)/\tau), \quad p_s^\tau(x') = \text{softmax}(f_s(x')/\tau) \quad (1)$$

by minimizing $\tau^2 D_{\text{KL}}(q_t^\tau(x) \| p_s^\tau(x'))$.

As τ increases, the output distributions become softer. After the usual τ^2 scaling, however, the distillation objective retains information about the relative structure of the underlying logits. Let $\mathbf{1} \in \mathbb{R}^C$ denote the all-ones vector and define the centered logits

$$\bar{f} = f - \frac{1}{C} \mathbf{1} \mathbf{1}^\top f,$$

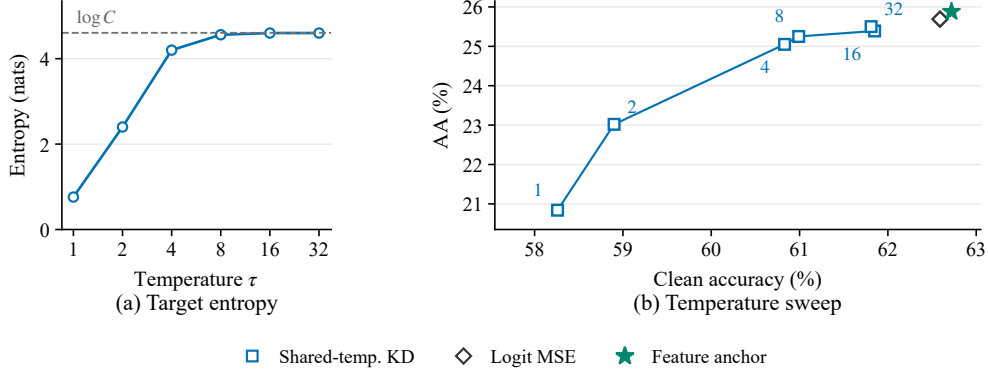


Figure 2: Target comparison on CIFAR-100 (ResNet-18). (a) Entropy of the clean teacher target as the shared temperature increases. (b) Clean accuracy and AutoAttack (AA) under shared-temperature knowledge distillation (KD), direct logit regression, and feature regression. Exact values are provided in Table 8.

which subtract the mean logit from every coordinate. For fixed teacher and student logits,

$$\lim_{\tau \rightarrow \infty} \tau^2 D_{\text{KL}}(q_t^\tau(x) \| p_s^\tau(x')) = \frac{1}{2C} \|\bar{f}_t(x) - \bar{f}_s(x')\|_2^2. \quad (2)$$

The centering reflects the invariance of softmax to adding the same constant to every logit. Thus, in the high-temperature limit, standard knowledge distillation approaches direct matching between the student’s perturbed output and a fixed clean teacher target, up to this shift invariance.

This establishes a connection between distribution matching and direct matching to a fixed clean target. CFA applies this fixed-target formulation to intermediate representations:

$$\mathcal{L}_{\text{anchor}}(x) = \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2. \quad (3)$$

Here, the clean teacher feature $\Phi_t(x)$ serves as a fixed target, and CFA trains the student to preserve this representation under perturbations within $B(x, \epsilon)$.

Figure 2 shows the corresponding empirical trend: as the temperature increases, KD approaches the operating point of direct regression, consistent with Equation (2). CFA uses the feature representation as its clean anchor and achieves a favorable clean–robust operating point. We next examine what controlling this anchor implies for the student.

3.2 WHAT THE ANCHOR CONTROLS

The fixed clean anchor provides a single reference for the student’s representations throughout the adversarial neighborhood. We now ask what is controlled when the worst-case discrepancy from this reference is kept small. For $B = B(x, \epsilon)$, define

$$L = \max_{x' \in B} \|\Phi_s(x') - \Phi_t(x)\|_2, \quad F = \|\Phi_s(x) - \Phi_t(x)\|_2, \quad O = \max_{x', x'' \in B} \|\Phi_s(x') - \Phi_s(x'')\|_2. \quad (4)$$

Here, F measures the student’s discrepancy from the teacher on the clean input, while O measures the variation of the student’s representation within the adversarial neighborhood.

Proposition 1 (Fixed-anchor decomposition). *For every x and $\epsilon \geq 0$,*

$$L \leq F + O \leq 3L. \quad (5)$$

The result follows directly from the triangle inequality. Since $x \in B(x, \epsilon)$, we have $F \leq L$. Moreover, for any $x', x'' \in B(x, \epsilon)$,

$$\|\Phi_s(x') - \Phi_s(x'')\|_2 \leq \|\Phi_s(x') - \Phi_t(x)\|_2 + \|\Phi_s(x'') - \Phi_t(x)\|_2 \leq 2L, \quad (6)$$

so $O \leq 2L$. Conversely, inserting $\Phi_s(x)$ between $\Phi_s(x')$ and $\Phi_t(x)$ gives $L \leq F + O$, completing the bound. The full proof and discussion are provided in Appendix A.

Thus, keeping the worst-case anchor discrepancy small simultaneously keeps the clean student representation close to the teacher and limits its variation within the perturbation region. Notably, this conclusion depends only on the teacher representation at the clean input; the teacher itself need not be robust throughout the neighborhood.

3.3 DOES BETTER CLEAN TEACHER PERFORMANCE TRANSFER TO THE STUDENT?

The fixed clean anchor raises a natural question: does a better clean teacher necessarily provide a better target for the adversarially trained student?

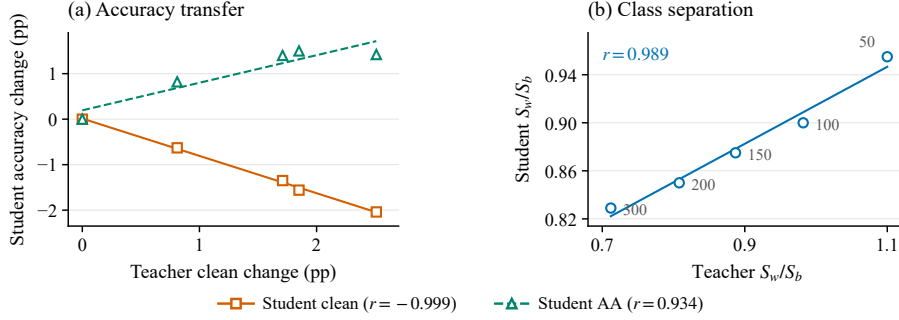


Figure 3: Transfer from clean self-teachers at different stages of training on CIFAR-100 (ResNet-18). (a) Changes in teacher clean accuracy and resulting student clean accuracy and AA relative to the 50-epoch teacher. (b) Ratio of within-class scatter S_w to between-class scatter S_b for teacher and student features. S_w measures the dispersion of features around their corresponding class centroids, whereas S_b measures the separation among class centroids; lower S_w/S_b indicates more compact within-class features relative to between-class separation. Labels denote teacher epochs, and lines are least-squares fits. Exact values are reported in Table 9.

As the clean teacher improves along its training trajectory, the resulting student does not exhibit a corresponding improvement in clean accuracy (Figure 3). Instead, different teacher checkpoints lead to different clean-robust operating points. The same overall trend appears for an independently trained teacher trajectory (Table 22).

We also examine the within-to-between scatter ratio S_w/S_b on unit-normalized clean features. Its evolution across teacher checkpoints is reflected in the resulting students, indicating that the transferred representation changes even when teacher quality is summarized only by clean accuracy. We treat this association as descriptive rather than causal.

A complementary comparison uses two self-teachers with nearly identical clean accuracy but different training schemes. Despite their similar clean performance, they produce substantially different student clean accuracies while yielding similar robustness (Table 1). Together, these results show that teacher clean accuracy alone is an incomplete descriptor of transferability. We therefore use the 200-epoch checkpoint as a fixed reference teacher in the remaining experiments.

Table 1: Transfer from accuracy-matched clean self-teachers on CIFAR-100 (ResNet-18; 50-epoch students). Feature sensitivity is measured on held-out data.

Natural training	Teacher		Student	
	Clean	Sensitivity	Clean	AA
Label smoothing 0.1	78.76	3.16	59.62	25.72
Mixup	78.62	4.82	63.80	25.70

3.4 SENSITIVITY TO INPUT PERTURBATIONS

The fixed clean target specifies what should be preserved, but the same input-space perturbation radius need not have the same effect on the anchoring objective for every training sample. For

$$\ell_{\text{anc}}(x'; x_i) = \|\Phi_s(x') - \Phi_t(x_i)\|_2^2,$$

the first-order change under an ℓ_∞ perturbation of radius e satisfies

$$\max_{\|\delta\|_\infty \leq e} \langle \nabla_{x'} \ell_{\text{anc}}, \delta \rangle = e \|\nabla_{x'} \ell_{\text{anc}}\|_1. \quad (7)$$

Thus, a common perturbation radius can induce different changes in the anchoring objective across training samples when their local sensitivities differ. This motivates adapting the perturbation radius to each sample’s sensitivity.

4 METHOD

The preceding analysis motivates two components of CFA: a fixed clean representation provides the target to preserve, while local sensitivity guides the perturbation radius assigned to each training sample.

4.1 CLEAN FEATURE ANCHORING

Let Φ_t, h_t denote the frozen clean self-teacher. The student is initialized from the teacher, $\Phi_s \leftarrow \Phi_t$ and $h_s = h_t$, after which only Φ_s is optimized. For each clean example,

$$x_{\text{adv}} = \arg \max_{x' \in B(x, \epsilon)} \|\Phi_s(x') - \Phi_t(x)\|_2^2, \quad \mathcal{L}_{\text{CFA}} = \mathbb{E}_x \|\Phi_s(x_{\text{adv}}) - \Phi_t(x)\|_2^2. \quad (8)$$

We approximate the inner maximization with randomly initialized PGD. Inference uses the inherited classifier $h_t \circ \Phi_s$. The full procedure is given in Algorithm 1.

4.2 SENSITIVITY-MATCHED ϵ

Motivated by the sensitivity analysis in Section 3.4, we adapt the perturbation radius to each training sample. More sensitive samples receive smaller radii, whereas less sensitive samples receive larger radii. Given a local sensitivity score $g_i > 0$ for sample i , we set

$$\epsilon_i = N \epsilon \frac{g_i^{-p}}{\sum_{j=1}^N g_j^{-p}}, \quad (9)$$

where N is the batch size, ϵ is the target mean radius, and p controls the strength of sensitivity matching. When $p = 0$, all samples receive the same radius ϵ ; for $p > 0$, the radius decreases with sensitivity. The normalization ensures $\frac{1}{N} \sum_{i=1}^N \epsilon_i = \epsilon$, so the average perturbation budget is unchanged.

For $p = 1$, the unnormalized rule satisfies $\epsilon_i g_i \propto 1$, approximately balancing the first-order perturbation effect across training samples. This relation motivates the allocation rule but does not imply exact equalization of the nonlinear anchoring loss.

We evaluate the effect of sensitivity matching and the importance of assigning the resulting radii to their corresponding samples in Tables 5 and 6. Details on sensitivity estimation, radius clipping, and sensitivity-norm choices are provided in Appendix F.

5 EXPERIMENTS

5.1 EXPERIMENTAL SETTINGS

We evaluate CFA on CIFAR-10, CIFAR-100 (Krizhevsky, 2009), and Tiny-ImageNet-200 (Le & Yang, 2015). We use ResNet-18 as the default architecture and additionally report results with WideResNet-34-10. Unless stated otherwise, the clean self-teacher is trained on natural images for

200 epochs and then frozen. The student is initialized from the teacher, with the inherited classifier kept fixed during adversarial training.

The main CIFAR experiments use 150 training epochs, AdamW with a one-cycle learning-rate schedule, 10-step projected gradient descent (PGD-10), $\epsilon_{\text{train}} = 8/255$, sensitivity-matched perturbation radii with $p = 1$, weight averaging (WA), and adversarial weight perturbation (AWP).

For comparison, we include standard adversarial training baselines (PGD-AT, TRADES, and MART), adversarial distillation methods (ARD, RSLAD, AdaAD, and AdaAD+IGDM), and methods that are particularly relevant to CFA’s clean-robust setting. The latter include LBGAT and ARREST, which use natural teachers; HAT and RPAT++, which target the clean-robust trade-off; and ADR, which uses self-distillation. We use the respective training recipes of each baseline, with the same natural teacher as CFA for distillation methods where applicable.

We report clean accuracy and AutoAttack (AA) accuracy on the full test set at $\epsilon = 8/255$, together with their arithmetic mean (Avg) and harmonic mean (NRR). Dataset-specific settings, teacher statistics, attack diagnostics, and additional controls are provided in Appendices G and J.

5.2 MAIN RESULTS

Table 2: ResNet-18 results under ℓ_∞ perturbations with $\epsilon = 8/255$. All values are measured in our framework. The CIFAR-100 CFA row reports the mean of three student seeds; all other rows are individual runs.

Method	CIFAR-10				CIFAR-100			
	Clean	AA	Avg	NRR	Clean	AA	Avg	NRR
PGD-AT	84.54	43.57	64.06	57.50	57.36	20.34	38.85	30.03
TRADES	81.71	49.16	65.44	61.39	55.28	23.55	39.42	33.03
MART	80.41	45.56	62.99	58.16	53.86	22.72	38.29	31.96
ARD	84.58	43.75	64.16	57.67	57.61	20.24	38.93	29.96
RSLAD	85.15	42.91	64.03	57.06	59.68	21.30	40.49	31.40
AdaAD	85.39	46.22	65.81	59.98	59.79	23.19	41.49	33.42
AdaAD + IGDM	85.01	45.50	65.26	59.27	48.25	19.48	33.87	27.75
LBGAT	—	—	—	—	56.46	26.02	41.24	35.62
ARREST	85.06	46.42	65.74	60.06	66.97	22.78	44.88	34.00
HAT	84.59	48.81	66.70	61.90	60.67	24.45	42.56	34.85
ADR	83.36	49.72	66.54	62.29	59.93	27.20	43.57	37.42
RPAT++	82.41	50.75	66.58	62.82	55.93	27.36	41.64	36.74
CFA (ours)	87.37	51.33	69.35	64.67	64.82	28.04	46.43	39.15

On CIFAR-100, CFA achieves 64.82% clean accuracy and 28.04% AA, with an NRR of 39.15. Compared with ADR, the closest baseline in NRR, CFA improves clean accuracy by 4.89 points, AA by 0.84 points, and NRR by 1.73 points. On CIFAR-10, CFA achieves 87.37% clean accuracy and 51.33% AA, with an NRR of 64.67, 1.85 points above the closest baseline (Table 2).

We additionally evaluate CFA on Tiny-ImageNet-200 and with WideResNet-34-10 to examine whether the results persist beyond the default CIFAR/ResNet-18 setting.

On Tiny-ImageNet-200 with ResNet-18, CFA achieves 55.16% clean accuracy and 20.54% AA, yielding an NRR of 29.93. With WideResNet-34-10 on CIFAR-10, CFA achieves 88.67% clean accuracy and 55.29% AA, with an NRR of 68.11 (Table 3). On CIFAR-100 with the same architecture, CFA obtains 65.04/30.14 clean/AA, compared with 64.50/29.04 for ADR (Table 19).

5.3 ABLATION AND COMPONENT ANALYSIS

Clean feature anchoring. We first isolate the effect of the anchor objective by removing sensitivity matching, weight averaging, and AWP. Under the same training recipe, CFA improves both clean accuracy and AA over label-based adversarial training (Table 23). The advantage remains when the cross-entropy baseline uses its own optimizer and training schedule.

Table 3: Results on Tiny-ImageNet-200 with ResNet-18 and CIFAR-10 with WideResNet-34-10, measured in our framework.

Method	Tiny-ImageNet, ResNet-18				CIFAR-10, WRN-34-10			
	Clean	AA	Avg	NRR	Clean	AA	Avg	NRR
PGD-AT	49.72	14.45	32.09	22.39	86.00	45.71	65.86	59.69
TRADES	46.98	15.95	31.46	23.83	85.13	48.99	67.06	62.19
MART	46.01	16.90	31.46	24.73	85.29	47.06	66.18	60.65
ARD	49.74	14.48	32.11	22.43	86.35	45.52	65.94	59.61
RSLAD	51.26	17.02	34.14	25.56	86.41	45.80	66.10	59.87
AdaAD	51.59	16.79	34.19	25.33	86.80	46.84	66.82	60.85
LBGAT	51.04	18.47	34.75	27.13	81.56	51.79	66.68	63.35
ARREST	59.38	16.06	37.72	25.28	87.54	48.83	68.19	62.69
HAT	52.57	17.28	34.93	26.01	87.15	53.21	70.18	66.07
ADR	47.66	20.85	34.25	29.01	88.64	52.34	70.49	65.82
CFA (ours)	55.16	20.54	37.85	29.93	88.67	55.29	71.98	68.11

Table 4: Feature geometry of CFA and label-based adversarial training on CIFAR-100. *Own* is the mean sample-to-own-centroid angle, and *Gap* is the mean angle from each class centroid to its nearest other-class centroid. Lower S_w/S_b and *Own* indicate tighter within-class concentration, whereas larger *Gap* indicates greater separation between class centroids. Angles are measured on unit-normalized features.

	Clean			Adversarial		
	S_w/S_b	Own	Gap	S_w/S_b	Own	Gap
CFA	0.709	26.3°	29.2°	1.969	33.5°	19.2°
Label CE	3.744	52.2°	33.5°	5.611	55.0°	28.2°

We further compare the resulting feature geometry. We measure the within-to-between class scatter ratio S_w/S_b , the mean angle of each feature to its own class centroid (*Own*), and the mean angle from each class centroid to its nearest other-class centroid (*Gap*).

As shown in Table 4, CFA produces substantially tighter within-class feature concentration on both clean and adversarial inputs, while the centroid gaps are not larger than those of the label-trained model. The difference is therefore characterized primarily by within-class concentration rather than increased separation between classes.

As an additional control, we apply teacher initialization, weight averaging, and AWP to the baseline methods. Although these additions improve most baselines, none reaches CFA’s NRR (Table 10).

Sensitivity-matched radius allocation. The allocation rule is motivated by the hypothesis that a common input-space radius does not correspond to a common perturbation effect under the anchor loss. We test this in two steps. We first compare sensitivity matching with a uniform radius while holding the teacher, training schedule, mean radius, WA, and AWP fixed. We then ask whether the sensitivity signal itself matters or whether heterogeneous radii alone are sufficient.

Table 5: Clean-target regression in the final CIFAR-100 recipe (ResNet-18, 150 epochs, $\epsilon_{\text{train}} = 8/255$). Feature rows report mean $\pm$ sample standard deviation over three student seeds and differ only in p ; logit regression is one run at $p = 1$.

Target and radius	Clean	AA	NRR
Features, uniform ($p = 0$)	62.63 ± 0.22	28.01 ± 0.20	38.71 ± 0.21
Features, sensitivity-matched ($p = 1$)	64.82 ± 0.07	28.04 ± 0.09	39.15 ± 0.09
Logits, sensitivity-matched ($p = 1$)	64.49	27.85	38.90

Across three paired runs, sensitivity matching improves clean accuracy by 2.19 ± 0.20 points while AA changes by only 0.03 ± 0.23 points. The clean gain appears in every seed, whereas the AA

difference is within run-to-run variation. Direct logit regression remains close to feature regression, consistent with the fixed-target interpretation.

Table 6: Radius-allocation signals under the same mean radius on CIFAR-100 with ResNet-18, trained for 100 epochs with WA and AWP. $\text{sd}(w)$ measures multiplier dispersion.

Allocation	$\text{sd}(w)$	Clean	AA	NRR
Uniform	0	60.42	28.42	38.66
Sensitivity	0.32	62.17	28.86	39.42
Same multipliers, difficulty order	0.32	60.90	28.06	38.42
Difficulty magnitude	0.39	61.27	28.12	38.55
Logit margin	0.42	61.07	27.79	38.20

Does the allocation signal matter? The “same multipliers, difficulty order” control is especially informative: it preserves the sensitivity-derived set of radii and changes only which examples receive them. This reassignment reduces AA by 0.80 points and NRR by 1.00 point. Thus the observed benefit cannot be attributed to radius heterogeneity alone; the correspondence between local sensitivity and the assigned neighborhood contributes to the result. Additional recipe, radius, and WA/AWP controls are reported in Tables 10, 15, 16 and 24.

6 CONCLUSION

We introduced CFA as a way to mitigate the clean–robust trade-off by separating the learning of clean knowledge from the learning of robustness. A naturally trained self-teacher first provides a high-accuracy clean reference; the student then learns robustness while matching that fixed reference over its adversarial neighborhood. Our analysis connects this view to fixed clean-target regression, characterizes the joint fidelity and local-variation control induced by the anchor, and shows that teacher clean accuracy alone does not determine robust transfer. Sensitivity-matched radii complement the anchor by redistributing a fixed perturbation budget according to the local response of the training objective. Across datasets, architectures, and controls, the results support clean self-teaching as a practical route to retaining clean performance during adversarial training without requiring an adversarial teacher or a label objective in the robust stage. The results also delineate the scope of this mechanism: fixed-target anchoring does not imply that feature coordinates are intrinsically preferable to logits, and clean teacher accuracy alone does not identify the most transferable teacher. Understanding which properties of a clean representation govern its transferability under adversarial training remains an open direction.

A PROOF AND SCOPE OF THE FIXED-ANCHOR DECOMPOSITION

Proof of Proposition 1. For any $x' \in B(x, \epsilon)$, the triangle inequality gives

$$\|\Phi_s(x') - \Phi_t(x)\|_2 \leq \|\Phi_s(x') - \Phi_s(x)\|_2 + \|\Phi_s(x) - \Phi_t(x)\|_2 \leq O + F,$$

so $L \leq F + O$. Since $x \in B(x, \epsilon)$, $F \leq L$. For any $x', x'' \in B(x, \epsilon)$,

$$\|\Phi_s(x') - \Phi_s(x'')\|_2 \leq \|\Phi_s(x') - \Phi_t(x)\|_2 + \|\Phi_s(x'') - \Phi_t(x)\|_2 \leq 2L,$$

so $O \leq 2L$ and therefore $F + O \leq 3L$. $\square$

The result concerns exact maxima over the perturbation set; finite-step PGD does not certify them. It applies to any fixed target in a normed space, including logits, and does not imply that features are superior to logits or that the bound alone guarantees robust classification.

B ALGORITHM

Algorithm 1 Clean Feature Anchoring (CFA)

```

1: Input: dataset  $\mathcal{D}$ , radius  $\epsilon$ , attack steps  $K$ , step size  $a$ , exponent  $p$ , clipping thresholds  $[w_{lo}, w_{hi}]$ 
2: Output: robust student  $h_t \circ \Phi_s$ 
3:
4: {Phase 1 — natural self-teacher. The only stage that uses labels.}
5: Train  $f_t = h_t \circ \Phi_t$  on  $\mathcal{D}$  with standard cross-entropy; freeze it
6:
7: {Phase 2 — anchored adversarial training. No labels, no head update.}
8:  $\Phi_s \leftarrow \Phi_t$  {student initialized at the teacher}
9: for epoch = 1 to  $T$  do
10:   for minibatch  $\{x_i\}_{i=1}^N \subset \mathcal{D}$  do
11:      $\phi_i \leftarrow \Phi_t(x_i)$  {teacher read once, at the clean input}
12:      $g_i \leftarrow \max(\|\nabla_x \|\Phi_s(x_i) - \phi_i\|_2, 10^{-12})$  {one backward pass}
13:      $v_i \leftarrow \text{clip}((\bar{g}/g_i)^p, w_{lo}, w_{hi}); w_i \leftarrow Nv_i / \sum_j v_j$ 
14:      $\epsilon_i \leftarrow \epsilon w_i; a_i \leftarrow aw_i$  { $\sum_i \epsilon_i = N\epsilon$ }
15:      $x_i^0 \leftarrow x_i + \eta_i$ 
16:     for  $k = 0$  to  $K - 1$  do
17:        $x_i^{k+1} \leftarrow \Pi_{B(x_i, \epsilon_i)}(x_i^k + a_i \text{sign } \nabla_x \|\Phi_s(x_i^k) - \phi_i\|_2)$ 
18:     end for
19:      $\mathcal{L} \leftarrow \frac{1}{N} \sum_i \|\Phi_s(x_i^K) - \phi_i\|_2^2$ 
20:      $\Phi_s \leftarrow \Phi_s - \mu \nabla_{\Phi_s} \mathcal{L}$  { $h_t$  is never updated}
21:   end for
22: end for

```

C TWO PARTIAL RESULTS ON THE ROBUSTNESS SIDE

Natural-teacher guidance is shared with prior work; our question is whether a fixed clean target can support adversarial training without an additional label objective. In CFA, the teacher specifies the target while the student’s inner maximization enforces matching under perturbations. Teacher vulnerability alone therefore does not determine the student’s response: matching a clean target is different from reproducing the teacher at perturbed inputs.

The two results below examine this distinction in restricted settings. The first studies robust and non-robust features in a two-feature model; the second relates anchor error to a classification margin. They supplement the fixed-anchor decomposition, but do not predict the measured accuracy or establish superiority over other regression targets.

C.1 IN THE TWO-FEATURE MODEL, THE NON-ROBUST COEFFICIENT IS ZEROED RATHER THAN SHRUNK

We use a restricted version of the feature model of Tsipras et al. (2018). One feature x_1 agrees with the label with probability $p = 0.95$; here, we assume it cannot be attacked. A further d features $z_j \sim \mathcal{N}(\eta y, 1)$ are individually almost uninformative, are jointly close to perfectly informative through their mean $m(z)$, and can all be moved by an ℓ_∞ adversary. We take $\Phi_t = m(z)$ as a natural teacher with near-perfect clean accuracy and poor robustness.

Write the student’s feature as $\Phi_s = ax_1 + bm(z)$. Then b is exactly the quantity the objection is about: how much of the teacher’s fragile route the student takes over. If the objection is right, minimizing our objective should leave b large.

Proposition 2. *With the inner maximum exact, $J(a, b) = \mathbb{E}[R^2] + 2\epsilon|b| \mathbb{E}|R| + \epsilon^2 b^2$ with $R = ax_1 + (b - 1)m$. Its one-sided slopes at $b = 0$ differ by $4\epsilon \mathbb{E}|R(a, 0)|$; above a threshold ϵ_0 , every minimizer has $b^* = 0$.*

Proof. The adversary reaches Φ_s only through m , and only through the mean perturbation $\bar{\delta} = \frac{1}{d} \sum_j \delta_j$, which ranges over $[-\epsilon, \epsilon]$ as $\|\delta\|_\infty \leq \epsilon$. The inner objective is therefore $\max_{|\bar{\delta}| \leq \epsilon} (R + b\bar{\delta})^2 = (|R| + \epsilon|b|)^2$, since a scalar quadratic is maximized at an endpoint and the sign of $\bar{\delta}$ is free. Taking expectations and expanding the square gives the display.

Fix a and differentiate in b . The term $\mathbb{E}[R^2]$ and $\epsilon^2 b^2$ are smooth, while $2\epsilon|b| \mathbb{E}|R|$ contributes $\pm 2\epsilon \mathbb{E}|R|$ as $b \rightarrow 0^\pm$. The one-sided derivatives at $b = 0$ are thus $2\epsilon \mathbb{E}[R(a, 0)m] \pm 2\epsilon \mathbb{E}|R(a, 0)|$, so $b = 0$ is a minimum along b exactly when

$$|\mathbb{E}[R(a, 0)m]| \leq \epsilon \mathbb{E}|R(a, 0)|. \quad (10)$$

At $b = 0$ the objective in a is $\mathbb{E}(ax_1 - m)^2$. Writing $q = 2p - 1$ for the mean label-alignment of x_1 , it is minimized at $a^* = \mathbb{E}[x_1 m] / \mathbb{E}[x_1^2] = q\eta$, and there $\mathbb{E}[R(a^*, 0)m] = q^2 \eta^2 - (\eta^2 + \frac{1}{d}) = -c$ with

$$c = \eta^2 v + \frac{1}{d}, \quad v = 1 - q^2. \quad (11)$$

Equation (10) therefore holds for every

$$\epsilon \geq \epsilon_0 = \frac{c}{\mathbb{E}|q\eta x_1 - m|}, \quad (12)$$

which is the stated threshold. For fixed x_1, m and $\bar{\delta} \in [-\epsilon, \epsilon]$, $(ax_1 + (b - 1)m + b\bar{\delta})^2$ is convex in (a, b) . Its maximum over $\bar{\delta}$ and then its expectation are convex, so J is jointly convex. At $(a^*, 0)$ the a -derivative vanishes, and Equation (10) puts zero in the b -subgradient; hence this point is a global minimizer. If $\epsilon > \epsilon_0$, the directional derivative from $(a^*, 0)$ is strictly positive in every direction with nonzero b . Along $b = 0$, $\mathbb{E}(ax_1 - m)^2$ has a unique minimizer. Thus every global minimizer has $b = 0$ above the threshold. $\square$

Why the coefficient is deleted rather than shrunk. The adversary contributes $2\epsilon|b| \mathbb{E}|R|$ to the loss — linear in $|b|$, not quadratic in b . A quadratic penalty pushes a coefficient toward zero and never arrives; a penalty on $|b|$ is kinked at the origin and sets the coefficient to zero outright once its slope dominates, which is the same distinction that makes the lasso sparse and the ridge not. What produces the kink here is that the student is asked to reproduce a *value* rather than a *route*: the fragile features carry that value on clean data but not inside the ball, so past a certain radius carrying any of them costs more than it pays. The teacher supplies what to match, and the inner maximization discards how the teacher got there.

At $d = 512$ the model’s constants give $c = 7.89 \times 10^{-3}$ and $\mathbb{E}|q\eta x_1 - m| = 0.0529$, hence $\epsilon_0 = 0.843 \eta$ — inside the bracket $(0.5\eta, 1.0\eta)$ that the Monte-Carlo minimization returns, and far below the 2η at which the model is usually stated.

The model assumes x_1 is unperturbable and z fully perturbable, so some suppression of z is built in. The proposition establishes the exact threshold behavior within this restricted model; it does not describe the geometry of real image classifiers.

That the exactness is the kink’s doing, and not a general consequence of adding an adversary, is visible from dropping it. With the $|b|$ term removed, $J_{\text{quad}} = \mathbb{E}[R^2] + \epsilon^2 b^2$; minimizing out a and writing $u = 1 - b$ collapses it to $u^2 c + \epsilon^2 (1 - u)^2$, so

$$b_{\text{quad}}^* = \frac{c}{c + \epsilon^2}, \quad (13)$$

which is strictly positive at every radius. Against the exact minimizer ($d = 512$, Monte-Carlo, $n = 2 \times 10^5$):

ϵ	b^* (exact)	b_{quad}^* (kink dropped)
0.5η	0.510	0.502
1.0η	0.000	0.202
2.0η	0.000	0.060
4.0η	0.000	0.016

The two columns agree below ϵ_0 and separate above it: the quadratic surrogate shrinks the non-robust coefficient and never deletes it, which is the ordinary picture, while the kinked objective sets it to zero and keeps it there.

Since every ϵ in J multiplies b , at $b^* = 0$ the objective reduces to $\mathbb{E}(ax_1 - m)^2$ and no longer depends on the radius. This saturation is a property of the two-feature model, not a prediction of the empirical radius sweep in Table 15.

The two-feature model cannot explain the measured clean–robust trade-off: it has only a robust and a fragile direction. It shows the narrower point that a robust student need not copy the fragile route by which its natural teacher computes the clean target.

C.2 WHAT ROBUST ACCURACY TRACKS IS THE LOSS MEASURED AGAINST A MARGIN

Writing $\gamma_t(x)$ for the teacher’s margin at x and D_y for the relevant decision-boundary constant, Cauchy–Schwarz gives that $\gamma_t(x) > D_y \cdot L(x)$ implies the student is robustly correct at x . The certificate is vacuous at the scales we train at, but the ratio DL/γ_t tracks both axes across the teacher ladder of Figure 3. What it does not do is predict the size of the effect, and it saturates before AutoAttack does.

D SUPPLEMENTARY ANALYSIS OF THE CLEAN-FEATURE ANCHOR

D.1 TEACHER EVALUATION ON PERTURBED INPUTS

The fixed-target comparison in Section 3.1 evaluates the teacher only at the clean input. This choice is important because the natural teacher is not locally robust: under $\epsilon = 8/255$, its feature rotates by 63.8° and its norm grows by a factor of 2.45. The main-paper control in Table 7 shows the consequence of replacing the fixed clean target with the teacher’s perturbed representation. Matching $\Phi_t(x_{\text{adv}})$ makes an exact copy of the vulnerable teacher a zero-loss solution and yields zero AutoAttack accuracy, whereas matching the fixed clean target retains nonzero robustness. These diagnostics reinforce that the teacher supplies a clean reference value rather than a robust perturbed response.

Table 7: Effect of the teacher input used to define the regression target on CIFAR-100 with ResNet-18. Both variants use the same 50-epoch diagnostic setting; only the teacher input used by the target and inner maximization differs.

Teacher target	Clean	AA	NRR
$\Phi_t(x)$	61.33	26.19	36.71
$\Phi_t(x_{\text{adv}})$	76.11	0.00	0.00

IGDM (Lee et al., 2025) uses the finite difference $f_t(x + \delta) - f_t(x - \delta)$ to approximate the teacher’s response along the perturbation direction. Its accuracy depends on local linearity. The reported

Taylor remainder proportion is 0.012 for adversarially trained teachers; we obtain 0.0111 for an adversarially trained ResNet-18 and 0.4763 for the natural teacher (Table 12). For the natural teacher, the finite difference has norm 14.14, compared with $\|f_t(x)\| = 18.36$, and its softmax assigns 0.917 probability to one class. These measurements indicate a less accurate local linear approximation and a highly concentrated target distribution.

We also evaluate published distillation objectives with the natural teacher replacing their robust teacher, using each method’s published recipe (Table 17). All evaluated objectives perform below adversarial training initialized from that teacher. This comparison complements the target sweep in Section 3.1, although differences among training recipes prevent it from isolating the target alone.

D.2 ADDITIONAL EVIDENCE FOR THE ANCHOR OBJECTIVE

We measure feature geometry on the final 150-epoch CFA model and a label-trained student with the same natural initialization, weight averaging, and AWP. Both models are evaluated on 5,000 test images under the same 10-step true-label PGD attack at 8/255; all angular statistics use unit-normalized features.

Proposition 1 bounds variation within an anchored model, but does not predict a smaller variation than label training. Under the matched true-label PGD protocol, the anchor and label-trained students move by 0.312 and 0.281 of their clean feature norms. These displacements are not the worst-case diameter O and provide no evidence of smaller relative displacement under anchoring. The reproducible difference is instead within-class concentration: CFA has a $5.3\times$ smaller clean S_w/S_b and a $2.85\times$ smaller ratio under attack (Table 4). Its class centroids are not farther apart than those learned with labels, so the result supports tighter clusters rather than greater inter-class separation.

The classification benefit is evaluated in Section 5.3. Without weight averaging, AWP, or the sensitivity-matched radius, the anchor improves clean and AutoAttack accuracy over label cross-entropy by 6.69 and 5.91 points under the same training conditions. The gains are 4.63 and 4.45 points when cross-entropy uses its own optimizer and schedule (Table 23). At the full training configuration, the AutoAttack gain over cross-entropy with weight averaging and AWP is 2.40 points. Under a matched attack, CFA has slightly larger relative feature displacement and smaller centroid gaps, but substantially tighter class-conditional features. This association does not establish concentration as the cause of the robustness gain.

The checkpoint ladder in Figure 3 further separates within-class concentration from between-class separation on unit-normalized features. From the 50-epoch to the 300-epoch teacher, the mean angle from a sample to its class centroid decreases from 34.6° to 30.9° , whereas the nearest-other-centroid angle between class means increases from 35.7° to 44.9° . The corresponding students reproduce the latter change—their nearest-centroid separation increases from 25.4° to 29.8° —while their mean within-class angle remains nearly unchanged (27.9° to 28.2°). These measurements motivate interpreting the ladder through between-class geometry, but they remain observational because the teacher’s accuracy and geometry change together. The numerical values underlying the two analysis figures are collected below.

The target comparison uses the 50-epoch base regime at $\epsilon = 8/255$, without weight averaging or AWP.

Table 8: Numerical values for the target comparison in Figure 2. $H(q_t^\tau)$ is measured in nats, and NRR is the harmonic mean of clean and AA accuracy.

Target	τ	$H(q_t^\tau)$	Clean	AA	NRR
<i>Teacher-only temperature</i>					
KL	1	0.76	58.26	20.84	30.70
KL	2	2.40	59.33	22.79	32.93
KL	4	4.20	59.39	24.48	34.67
KL	8	4.56	59.47	25.19	35.39
KL	16	4.60	57.78	24.00	33.91
<i>Shared-temperature KD</i>					
KL	1	0.76	58.26	20.84	30.70
KL	2	2.40	58.90	23.02	33.10
KL	4	4.20	60.83	25.05	35.49
KL	8	4.56	60.99	25.25	35.71
KL	16	4.60	61.85	25.39	36.00
KL	32	4.60	61.81	25.50	36.11
Logit regression, fixed head	—	—	62.59	25.69	36.43
Feature anchor, fixed head	—	—	62.72	25.88	36.64

Table 9: Numerical values for the checkpoint trends in Figure 3.

Checkpoint	Teacher		Student		
	Clean	S_w/S_b	Clean	S_w/S_b	AA
50 epochs	75.81	1.100	64.28	0.955	24.38
100 epochs	76.62	0.982	63.65	0.900	25.20
150 epochs	77.52	0.887	62.93	0.875	25.78
200 epochs	77.66	0.808	62.72	0.850	25.88
300 epochs	78.32	0.712	62.24	0.829	25.80

E ADDITIONAL RESULTS

Baseline training-stack control. To test whether CFA’s final performance can be explained by teacher initialization, weight averaging, and AWP alone, we add the same components to the baseline recipes. Most baselines improve, but none reaches CFA’s NRR.

Table 10: Adding teacher initialization, weight averaging, and AWP to baseline recipes on CIFAR-100 (ResNet-18).

Method	Own recipe			+ teacher init + WA + AWP		
	Clean	AA	NRR	Clean	AA	NRR
PGD-AT	57.36	20.34	30.03	59.85	25.77	36.03
TRADES	55.28	23.55	33.03	56.16	24.56	34.17
MART	53.86	22.72	31.96	50.97	25.13	33.66
ARD	57.61	20.24	29.96	60.19	26.10	36.41
RSLAD	59.68	21.30	31.40	63.44	25.55	36.43
AdaAD	59.79	23.19	33.42	57.21	27.10	36.78
AdaAD + IGDM	48.25	19.48	27.75	54.24	25.56	34.75
Consistency-AT	57.53	20.01	29.69	60.74	25.38	35.80
LBGAT	56.46	26.02	35.62	53.91	25.16	34.31
ADR	59.93	27.20	37.42	59.84	27.84	38.00
HAT	60.67	24.45	34.85	57.63	24.74	34.62
ARREST	66.44	22.99	34.16	64.95	23.02	33.99
CFA (ours)	—			64.82	28.04	39.15

The following tables provide additional results supporting the main experiments and supplementary analysis.

Regression under weight averaging and AWP. Table 11 compares logit and feature regression with the same natural teacher, inherited frozen classifier, AdamW optimizer, one-cycle schedule, WA, and AWP. Both use a uniform radius: neither run includes sensitivity-based allocation. Each

objective is maximized to generate training perturbations and minimized to update the backbone, so the comparison includes its effect on both stages. Logit regression remains competitive with the training stack; the single-seed difference does not establish a general advantage for feature regression.

Table 11: Fixed-head regression with WA and AWP: CIFAR-100, ResNet-18, 100 epochs, uniform $\epsilon_{\text{train}} = 8.8/255$. Evaluation uses AA at $8/255$. Both losses sum squared errors across output coordinates and average over the batch.

Target	Clean	AA
Clean logits	60.36	28.02
Clean features	60.42	28.42

Table 12: Local linearity, which IGDM’s construction requires of the teacher: remainder proportion of the first-order Taylor expansion over an $\epsilon = 8/255$ ball, computed as in IGDM’s Sec. 3.1.

Network	Remainder proportion
IGDM’s robust teachers (LTD / BDM-AT / IKL-AT, published)	0.012 / 0.012 / 0.016
PGD-AT ResNet-18 (ours)	0.0111
Natural teacher (ours)	0.4763
Anchored student, $p = 0$	0.0029
Anchored student, reported recipe	0.0202

Table 13: What to do with the classifier. CIFAR-100 base regime (50 epochs, raw features, no weight averaging, no AWP, $\epsilon = 8/255$), so that no stack component can absorb the difference.

	Clean	PGD-20	CW	AA	NRR
Logit target only, $\tau = 1$	58.26	22.62	22.79	20.84	30.70
Logit target only, $\tau = 4$	59.39	30.30	26.84	24.48	34.67
Logit target only, $\tau = 16$	57.78	31.57	26.11	24.00	33.91
Feature anchor + head KD, $\tau = 1$	62.34	26.34	26.11	23.93	34.58
Feature anchor + head KD, $\tau = 16$	62.61	32.47	27.32	25.61	36.35
Feature anchor, head untouched	62.72	28.69	27.80	25.88	36.64
head refit, adversarial CE	62.77	29.93	27.66	25.65	36.42
head refit, adv. CE + smoothing 0.1	62.57	30.43	27.55	25.66	36.39
head refit, clean CE	62.70	28.90	27.23	25.15	35.90

Normalization check. Because Section 3.3 analyzes angular geometry, we check whether discarding feature magnitude changes the training conclusion. Normalizing both sides changes NRR by 0.18 with weight averaging and AWP, while raw regression is better in the base regime. We therefore retain raw squared ℓ_2 as the default and use normalized features only for analysis; CFA does not rely on a cosine objective.

Table 14: Effect of feature normalization in the anchor loss. Normalizing both representations to the unit sphere removes their magnitudes and keeps only the angle, for which the squared Euclidean loss satisfies $\|\hat{\Phi}_s - \hat{\Phi}_t\|_2^2 = 2 - 2 \cos \angle(\Phi_s, \Phi_t)$ and is therefore equivalent to cosine loss up to an affine transformation. CIFAR-100, ResNet-18, 100 epochs; within each panel, the runs differ only in how the student and teacher features are normalized. The asymmetric row normalizes the student alone, placing the two sides at different scales, and is included only as a diagnostic.

	Clean	AA	NRR
<i>(a) With the stack: WA, AWP, $\epsilon_{\text{train}} = 8.8/255$</i>			
Raw features (squared ℓ_2)	60.21	28.14	38.35
Both normalized (cosine)	61.41	28.07	38.53
<i>(b) Base regime: no WA, no AWP, $\epsilon_{\text{train}} = 8/255$</i>			
Raw features (squared ℓ_2)	62.40	24.34	35.02
Both normalized (cosine)	61.52	22.90	33.38
Asymmetric: student normalized, teacher raw	61.28	22.98	33.43

Table 15: ϵ_{train} swept with everything else fixed. Evaluation is at $\epsilon = 8/255$ in every cell.

	Clean	PGD-20	CW	AA	NRR
<i>CIFAR-100</i>					
6/255	68.07	28.40	28.16	25.82	37.44
7/255	65.61	30.34	29.64	27.44	38.70
8/255	63.89	31.41	29.83	27.78	38.72
8.8/255 [†]	62.17	32.37	30.93	28.86	39.42
10/255 [†]	59.99	32.82	30.53	28.77	38.89
<i>CIFAR-10</i>					
6/255	89.64	49.29	50.20	47.88	62.42
7/255	88.30	51.43	52.64	50.27	64.07
8/255 [†]	87.22	52.43	53.55	51.15	64.48
8.8/255 [†]	84.96	52.98	53.94	51.74	64.31
10/255 [†]	83.29	53.25	53.88	51.79	63.87

Table 16: AWP on top of weight averaging, across both datasets and both training radii. Every run uses the sensitivity-matched radius for 100 epochs; the AWP proxy uses $\gamma = 0.005$ after a 10-epoch warmup, and within each row AWP is the only added component. This is the right-hand edge of the factorial in Table 24, repeated on CIFAR-10 to check that the sign of the AWP effect does not depend on the dataset.

	WA			WA + AWP		
	Clean	AA	NRR	Clean	AA	NRR
<i>CIFAR-100</i>						
8/255	63.54	26.75	37.65	63.89	27.78	38.72
8.8/255	62.30	27.69	38.34	62.17	28.86	39.42
<i>CIFAR-10</i>						
8/255	86.67	49.20	62.77	87.22	51.15	64.48
8.8/255	85.16	50.15	63.13	84.96	51.74	64.31

Table 17: Published adversarial-distillation objectives supplied with our natural teacher. The first row of each method uses its published recipe; indented rows add our teacher initialization and then without weight averaging or AWP.

	CIFAR-100			CIFAR-10		
	Clean	AA	NRR	Clean	AA	NRR
ARD	57.61	20.24	29.96	84.58	43.75	57.67
+ teacher initialization	57.96	20.22	29.98	—	—	—
RSLAD	59.68	21.30	31.40	85.15	42.91	57.06
+ teacher initialization	59.87	20.71	30.77	—	—	—
AdaAD	59.79	23.19	33.42	85.39	46.22	59.98
+ teacher initialization	59.75	22.78	32.98	—	—	—
AdaAD + IGDM	48.25	19.48	27.75	85.01	45.50	59.27
+ teacher initialization	49.67	20.15	28.67	—	—	—
PGD-AT, teacher initialization	57.73	26.46	36.29	82.23	50.72	62.74
CFA, no WA/AWP	61.29	25.36	35.88	—	—	—

Table 18: RPAT++ given the rest of our stack. Its own recipe already carries weight averaging (EMA 0.999 from epoch 105 of 200), so the missing component is the AWP proxy, added at $\gamma = 0.005$ with warmup at 10% of the run and evaluated on its own loss; the last row also starts the run at our natural teacher. CIFAR-100, ResNet-18, our reproduction, reported at both the final and the PGD-20-selected weight-averaged checkpoint.

	Final			Best by PGD-20		
	Clean	AA	NRR	Clean	AA	NRR
RPAT++ (its own recipe)	57.05	26.61	36.30	55.93	27.37	36.74
+ AWP	57.48	28.05	37.70	57.50	27.92	37.60
+ AWP + teacher initialization	57.83	27.47	37.25	57.80	27.59	37.36

F THE ALLOCATION RULE AND ITS BUDGET SOLVE

F.1 WHERE THE RULE COMES FROM

The objective is optimized over a **pixel** ball, while what it measures is a displacement in feature space, and a fixed pixel radius moves that displacement by very different amounts across samples. The allocation follows from one elementary fact: for a differentiable loss the first-order movement available inside an ℓ_∞ ball of radius e is

$$\max_{\|\delta\|_\infty \leq e} \langle \nabla_x \mathcal{L}, \delta \rangle = e \|\nabla_x \mathcal{L}\|_1, \quad (14)$$

the dual norm of ℓ_∞ being ℓ_1 . Equalizing that movement across a batch subject to a fixed total budget $\sum_i \epsilon_i = N\epsilon$ gives $\epsilon_i g_i = \text{const}$, that is $\epsilon_i \propto 1/g_i$ with $g_i = \|\nabla_x \mathcal{L}(x_i)\|_1$. The same allocation is the max-min one: if two samples move by different amounts, shifting budget from the sample that moves more to the sample that moves less raises the minimum, so no unequal assignment is optimal. Taking $\epsilon_i \propto (\bar{g}/g_i)^p$ makes $p = 1$ that allocation, $p = 0$ the uniform-pixel one, and $p \in (0, 1)$ an interpolation.

Two approximations are worth declaring rather than leaving to be found. First, g_i is measured once, by a single backward pass at the *clean* x_i before the attack starts, so it is the sensitivity at the starting point and not along the PGD trajectory. Second, everything above is first order in ϵ . Neither is repaired anywhere in the method; the rule is a budget heuristic with a derivation, not an exactly solved subproblem.

We also do not claim the rule is specific to a directional objective. $\|\nabla_x \mathcal{L}\|_1$ is defined for any loss and the implementation reads whichever loss is configured; on the shipped raw- ℓ_2 anchor it equalizes a feature displacement rather than an angle, which is why the honest name is *sensitivity-matched* ϵ rather than an angular budget. What separates it from IAAT, MMA and CAT is not directionality but the signal: they allocate from a property of the sample, this allocates from the geometry the training loss is written in.

The separation is not one of dispersion, which is worth stating because a rule that simply spread the budget more aggressively would be the simpler explanation of any gain. Measured over training on the reported configuration, the coefficient of variation of our sensitivity signal has median 0.56, against 0.75 for the difficulty score IAAT and CAT allocate from and 1.33 for the logit margin of MMA. The box of this section compresses all three before they reach the attack, leaving per-sample multipliers of spread 0.32, 0.39 and 0.42 respectively, and the ordering survives the compression. Ours is the least dispersed of the three signals either way, so the comparison in Table 6 is a comparison of which geometry the budget follows.

The first-order derivation and the reported implementation use different sensitivity norms. Equation (14) gives the ℓ_1 norm and the reported implementation uses the ℓ_2 gradient norm, which was fixed before this comparison and retained for reproducibility. The two are not the same allocation. Measured over training on the reported configuration, the ℓ_1 signal’s coefficient of variation has median 0.68 against the ℓ_2 signal’s 0.56, and the per-sample multiplier the attack receives has spread 0.34 against 0.32 — so the norms really do distribute the budget differently.

They nonetheless land close together. Training the reported recipe with `featdir_angepts_gnorm` set to 1 and nothing else changed gives 62.16/31.97/30.56/28.57 (NRR 39.15) against 62.17/32.37/30.93/28.86 (NRR 39.42): clean accuracy is identical to a hundredth and AutoAttack differs by 0.29, in favour of the norm we ship.

The difference is small but non-negligible relative to the observed seed variation. It is twice the seed-to-seed spread of Appendix G, and it is the same order of magnitude as what the allocation buys in the first place: against the uniform radius, $p = 1$ gains +1.75 clean and +0.44 AutoAttack at ℓ_2 and +1.74 and +0.15 at ℓ_1 . So the rule’s benefit survives the norm in direction and almost entirely on the clean axis, while most of its AutoAttack half does not. Thus the sensitivity norm is a genuine implementation choice: the reported ℓ_2 variant performs slightly better in AA, while the clean-accuracy gain persists with the theoretically matched ℓ_1 norm.

The permuted control in Table 6 isolates the assignment signal from the radius distribution. It takes the multiset of weights the gradient rule produces — the same values, the same clip $[0.5, 1.5]$, the same restored mean — and only permutes which example receives which, so that the larger radii go to the easier examples, which is the direction IAAT and CAT assign them. Reordering costs 0.80 AutoAttack against the rule it reorders and lands below even the uniform row on AutoAttack, CW and NRR alike.

F.2 CLIPPING AND RESTORING THE MEAN RADIUS

The reported configuration floors the gradient norm at 10^{-12} and computes $v_i = \text{clip}((\bar{g}/g_i)^p, 0.5, 1.5)$, followed by $w_i = Nv_i / \sum_j v_j$ and $\epsilon_i = \epsilon w_i$. PGD uses step size $a_i = aw_i$. This preserves the batch mean radius exactly. The clipping thresholds apply before normalization, so they are not hard bounds on the final multipliers. When all gradients vanish, the floor makes all g_i equal and the allocation uniform.

Optional exact-box variant. The implementation also supports an alternative that constrains the final multipliers. It is not enabled in the reported default configuration.

With $r_i = (\bar{g}/g_i)^p$ and box $[w_{\text{lo}}, w_{\text{hi}}] = [0.5, 1.5]$, the radii that satisfy both the box and the budget exactly are $\epsilon_i = \epsilon \cdot \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}})$ with t the solution of

$$\sum_{i=1}^N \text{clip}(tr_i, w_{\text{lo}}, w_{\text{hi}}) = N. \quad (15)$$

The left side is continuous and nondecreasing in t from Nw_{lo} to Nw_{hi} , so a root exists and bisection finds it; when no clip is active it reduces to $t = N / \sum_i r_i$, that is, to restoring the mean. Clipping first and rescaling afterwards, the obvious alternative, does not preserve the box—the rescaling moves the clipped entries back out of it.

Table 19: WideResNet-34-10 on CIFAR-100 at $\epsilon = 8/255$, measured in our framework.

Method	Clean	AA	Avg	NRR
PGD-AT	58.97	22.84	40.91	32.93
TRADES	57.35	25.60	41.48	35.40
MART	57.16	22.43	39.80	32.22
ARD	59.52	23.05	41.29	33.25
RSLAD	60.48	21.63	41.06	31.87
AdaAD	61.66	25.62	43.64	36.20
LBGAT	59.63	28.55	44.09	38.61
ARREST	67.69	24.69	46.19	36.18
HAT	62.81	26.91	44.86	37.68
ADR	64.50	29.04	46.77	40.05
CFA (ours)	65.04	30.14	47.59	41.19

G EXPERIMENTAL DETAILS

Table 20: Natural teacher checkpoints used by the shipped models. The teachers are trained only on clean examples and have zero adversarial accuracy.

Dataset	Architecture	Epochs	Clean
CIFAR-10	ResNet-18	200	95.32
CIFAR-100	ResNet-18	200	77.66
Tiny-ImageNet	ResNet-18	200	66.29

Every number we report is the **final** epoch, never a checkpoint selected on a robustness metric. This matters when reading Table 27: RPAT selects on PGD-20 and ADR on PGD-10, so those rows are upper envelopes of a training run where ours is its endpoint.

Teacher. One naturally trained network per dataset, no adversarial examples anywhere in its training, and the student is warm-started from the same checkpoint. ResNet-18, SGD at learning rate 0.1 under a one-cycle schedule, batch size 128, no augmentation beyond the standard crop and flip, for 200 epochs on each dataset. The resulting teachers are summarized in Table 20. The teacher’s own robustness is 0 — it is never attacked and is not meant to survive attack; Figure 3 is the ablation over how long it is trained.

Student. The main ResNet-18 configurations are listed below. Unless noted otherwise, CIFAR-100 ablations use 100 epochs at $8.8/255$, with a 10-epoch AWP warmup.

	CIFAR-10	CIFAR-100	Tiny-ImageNet-200
Backbone	ResNet-18	ResNet-18	ResNet-18
Optimizer	AdamW	AdamW	AdamW
Learning rate / schedule	0.021, one-cycle	0.021, one-cycle	0.021, one-cycle
Epochs	150	150	100
Batch size	128	128	128
Training radius ϵ_{train}	$8/255$	$8/255$	$8.8/255$
Inner attack	PGD-10, step 2/255	PGD-10, step 2/255	PGD-10, step 2/255
Allocation exponent p	1	1	1
Radius box $[w_{\text{lo}}, w_{\text{hi}}]$	$[0.5, 1.5]$	$[0.5, 1.5]$	$[0.5, 1.5]$
Classifier head	frozen	frozen	frozen
Weight averaging (start, decay)	0.2, 0.999	0.2, 0.999	0.2, 0.999
AWP (γ , warmup)	0.005, 15 ep	0.005, 15 ep	0.005, 10 ep
Teacher checkpoint	clean, 200 ep	clean, 200 ep	clean, 200 ep

We keep the inherited classifier fixed and omit the head-KD term, removing its temperature τ and weight β from the objective. p is not tuned. The training-radius sweep is reported in Table 15; evaluation is at $8/255$ in every cell.

Evaluation. We report AutoAttack accuracy at ℓ_∞ radius 8/255 on the full test set; NRR is its harmonic mean with clean accuracy. Additional attacks are in Appendices J and L. Ported baselines retain their published recipes unless stated otherwise.

Repeated training. We repeat both arms of the final radius-allocation comparison with three student seeds. Uniform radii give 62.63 ± 0.22 clean and 28.01 ± 0.20 AA; sensitivity matching gives 64.82 ± 0.07 clean and 28.04 ± 0.09 AA (mean $\pm$ sample standard deviation). The paired clean gains are 2.35, 2.24, and 1.97 points, whereas the paired AA changes are 0.14, -0.23 , and 0.18. Thus the clean gain repeats, while the data support no AA change. Other rows remain individual runs, so small differences between regression targets or nearby operating points should not be interpreted as significant.

Compute, teacher included. A single RTX 5080. One student cell is about 2.2 hours on CIFAR-100 at 100 epochs, of which the AutoAttack evaluation is roughly 7 minutes; Tiny-ImageNet is longer in proportion to its test set.

The teacher has to be counted, since calling it free would otherwise hide a training run. It is not free, but it is cheap: a clean epoch has no inner maximization, so it costs 6.8 seconds against 74.5 for an adversarial one, and 200 clean epochs come to 0.39 hours. The teacher is therefore 19% of one student run and is amortized over every cell that reads it, the teacher ladder of Figure 3 included. Table 21 puts the total against the baselines measured on the same machine.

Table 21: Wall-clock training cost on one RTX 5080, CIFAR-100, ResNet-18, as logged by the runs reported in this paper. Evaluation is excluded. Adversarial distillation additionally requires an adversarially trained teacher, which is not counted in its row and is the more expensive artifact.

Method	Epochs	Teacher (h)	Student (h)	Total (h)
Natural teacher (ours, reused)	200 clean	–	0.39	0.39
CFA (ours)	100 adv.	0.39	2.07	2.46
Label CE at our regime	100 adv.	0.39	3.22	3.61
PGD-AT, initialized at the teacher	100 adv.	0.39	1.72	2.11
ARD	100 adv.	0.39	1.52	1.91
RSLAD	100 adv.	0.39	1.71	2.10
AdaAD	100 adv.	0.39	2.79	3.18
LBGAT	100 adv.	joint	1.88	1.88
HAT	50 adv.	0.39	1.10	1.49
ADR	200 adv.	self	3.52	3.52

Two readings follow. The anchor is not bought with compute, since it costs less end to end than label cross-entropy at the same regime and about what PGD-AT costs once the teacher is included. The adversarial-distillation rows additionally rely on robust teachers with different training requirements, so their total teacher–student cost is not directly comparable to CFA’s natural self-teacher setup.

H ADDITIONAL TEACHER-CHECKPOINT RESULTS

Figure 3 shows one natural-training trajectory; Table 22 reports an independently trained trajectory. Each listed teacher was trained to completion under its own one-cycle schedule, then used to initialize a 50-epoch student with the same training procedure. These experiments vary the teacher checkpoint, not just the target features.

The second trajectory again moves from higher student clean accuracy at earlier checkpoints to higher AA at later ones. The strongest AA occurs at 200 rather than 300 epochs, although the difference is small. Feature sensitivity has only moderate rank correlation with student clean accuracy on these six checkpoints (Spearman 0.54, versus 0.83 on the same grid of the first trajectory). The matched teachers in Table 1 further show that teacher clean accuracy alone does not determine student clean accuracy. None of these measurements supplies a validated teacher-only selection rule.

Table 22: Independent CIFAR-100 teacher trajectory (teacher seed 1; 50-epoch students with seed 0). Teacher metrics are measured on held-out data.

Epoch	Teacher			Student		
	Clean	Margin	Sensitivity	Clean	AA	NRR
20	73.87	3.074	3.911	63.82	23.22	34.05
40	75.57	3.931	4.542	64.36	23.86	34.81
100	77.27	4.506	4.678	63.68	25.61	36.53
150	77.68	4.495	4.262	63.37	25.49	36.36
200	77.90	4.542	4.129	63.02	26.16	36.97
300	78.27	4.523	3.835	62.34	25.94	36.64

I THE RECIPE CONTROLS

Unless explicitly ablated, CFA warm-starts from the natural teacher and trains with AdamW at 0.021 under a one-cycle schedule, while the baselines retain their authors’ recipes. Table 23 asks whether that recipe is what wins: it runs our own objective, and label cross-entropy for reference, under both recipes and both initializations. Every row is measured here at the base regime, which carries no weight averaging, no AWP and a uniform radius, on CIFAR-100 with ResNet-18 for 100 epochs.

Table 23: The recipe controls. *Ours* is AdamW at 0.021 under a one-cycle schedule with the teacher warm start; *theirs* is SGD at 0.1 decayed by ten at epochs 70 and 90, the schedule PGD-AT uses here. The training radius and every other field are held fixed within each pair.

Objective	Recipe	Clean	AA	Avg	NRR
Anchor	ours, teacher init	61.29	25.36	43.33	35.88
Anchor	ours, random init	60.81	25.21	43.01	35.66
Anchor	theirs, teacher init	59.65	25.54	42.60	35.77
Label cross-entropy	ours, teacher init	54.60	19.45	37.03	28.68
Label cross-entropy	theirs, teacher init	56.66	20.91	38.79	30.55

Changing initialization or schedule changes the anchor’s NRR by 0.22 and 0.11, respectively. Cross-entropy improves by 1.87 NRR under its own schedule, and that stronger baseline is the one carried into the comparisons in the main text. The anchor remains ahead under both tested schedules. These comparisons support an objective-level contribution within the tested recipes, without establishing significance from single runs or claiming exhaustive tuning of the alternatives.

The ladder in Table 24 is reported as a factorial rather than as one ordered sequence, because the order of addition is not identifiable from a single chain. Weight averaging and AWP are crossed with each other on top of the same anchor and the same sensitivity-matched radius, and the factorial is repeated at both training radii so that the pattern can be separated from the radius choice.

Table 24: Weight averaging crossed with AWP on the CIFAR-100 ResNet-18 anchor (100 epochs, the sensitivity-matched radius with $p = 1$ in every cell, evaluated at $\epsilon = 8/255$). The two panels repeat the same 2×2 design at the two training radii. The top-left cell of each panel is the anchor with no stack component.

WA	AWP	Clean	AA	Avg	NRR
$\epsilon_{\text{train}} = 8/255$					
–	–	64.11	24.90	44.50	35.87
–	✓	65.01	27.62	46.32	38.77
✓	–	63.54	26.75	45.14	37.65
✓	✓	63.89	27.78	45.84	38.72
$\epsilon_{\text{train}} = 8.8/255$					
–	–	63.09	25.74	44.41	36.56
–	✓	63.11	27.97	45.54	38.76
✓	–	62.30	27.69	44.99	38.34
✓	✓	62.17	28.86	45.52	39.42

Both components raise AA on their own, and their effects on AA are sub-additive. At 8/255 AWP alone is worth +2.72 AA and +2.90 NRR over the bare anchor, while weight averaging alone is worth +1.85 AA and +1.78 NRR; adding weight averaging on top of AWP then returns only +0.16 AA, and its clean cost of 1.12 points leaves NRR flat (-0.05). The same ordering holds at 8.8/255, where AWP alone gives +2.20 NRR and weight averaging alone +1.78, but the second component adds only +0.66 and +1.08 NRR respectively. The two components also differ in sign on clean accuracy: AWP never costs clean accuracy here and at 8/255 gains 0.90 points of it, whereas weight averaging costs between 0.57 and 1.12 clean points in all four of its cells.

Two readings are consistent with this table — much of what the stack contributes is available from AWP alone, and the two components overlap in what they contribute — and we do not try to separate them further. We keep the full stack as the reported recipe because it gives the best AA in both panels and the best NRR at 8.8/255; at 8/255 AWP alone is ahead of it by 0.05 NRR, which is smaller than the seed-to-seed spread we measure at the reported configuration. Either choice leaves the comparisons in Table 2 unaffected, since every CFA row and every stack-matched baseline row carries the same two components.

J GRADIENT MASKING

We test for gradient masking with AutoAttack, independent attacks, and PGD sweeps (Athalye et al., 2018). Table 25 reports these checks for a 100-epoch CIFAR-100 repeat with 62.00% clean accuracy.

Table 25: Gradient-masking checks on all 10,000 CIFAR-100 test images at 8/255. Individual attacks are evaluated independently.

Check	Setting	Accuracy
<i>white box against black box</i>		
APGD-CE	white box, untargeted	31.80
APGD-T	white box, targeted	28.83
FAB-T	white box, boundary	29.16
Square	black box, 5000 queries	34.82
AutoAttack	standard evaluation	28.73
<i>radius sweep, PGD-20</i>		
$\epsilon = 2/255$		54.62
$\epsilon = 4/255$		47.03
$\epsilon = 8/255$		32.46
$\epsilon = 12/255$		20.88
$\epsilon = 16/255$		11.94
$\epsilon = 32/255$		1.05
$\epsilon = 64/255$		0.00
<i>step sweep, $\epsilon = 8/255$</i>		
PGD-10 / 20 / 50 / 100		33.17 / 32.57 / 32.30 / 32.24
<i>transfer</i>		
Natural teacher $\rightarrow$ student	PGD-20 built on the teacher	60.42

AutoAttack yields 28.73% accuracy. PGD accuracy falls monotonically with attack radius, reaching zero at 64/255, and decreases with additional steps. These checks show no indication of gradient masking under the evaluated attacks.

The transfer row is worth reading separately. The student inherits the teacher’s classifier and is initialized at the teacher’s features, so adversarial examples built on the teacher might be expected to carry over; they cost the student 1.58 points against its clean accuracy. Sharing a head is therefore not sharing a vulnerability, which is the same conclusion Section 5.3 reaches from class separation: what the anchored student changes is the position of its features, and the teacher’s own adversarial directions do not point at the student’s boundaries.

K NOTES ON THE BASELINES

Several rows of Table 2 run a specific variant of the method they name, or are grouped in a way that needs saying. This section is those decisions; the table carries none of them.

ARREST uses its published objective: adversarial finetuning from the natural model with SGD at 0.025, the angular representation term at $\lambda = 50$, and Noisy Replay over the first half at $\phi = 30^\circ$. Table 2 reports its 150-epoch run for the main comparison, whereas Table 10 uses its published 20-epoch schedule. We read its representation after global pooling, where our anchor and geometry measurements are defined.

LBGAT, and why its CIFAR-10 cell is empty. The authors release a training script for CIFAR-100 only, and their CIFAR-10 entries are released as pretrained WideResNet checkpoints. Running their CIFAR-100 recipe on CIFAR-10 diverges here within thirty steps: the squared error is reduced with mean over batch and classes, so a ten-class problem receives ten times the gradient of a hundred-class one at equal logit error, and $\text{lr} = 0.1$ is above the stability threshold that follows. Lowering the learning rate trains (at 0.01 the natural branch reaches 49% within 180 steps), but a learning rate we chose is not their recipe, so the cell is left empty rather than filled with a number of our own construction. LBGAT trains its natural branch jointly with the robust one, so unlike the distillation rows its teacher is not obtained for free, and we group it accordingly. The two datasets run different variants because the authors publish different ones: CIFAR-100 is LBGAT6 and CIFAR-10 is LBGAT0, their release lists CIFAR-10 checkpoints under LBGAT0 alone, and $\beta = 6$ is their CIFAR-100 setting. We report the configuration each dataset has a published counterpart for rather than forcing one β across both.

HAT uses a naturally trained network only to *label* helper points beyond the ball, never as a target, so it belongs with the trade-off methods and not the distillation ones. It is run on its own one-cycle schedule rather than ours, which collapses it to chance on both datasets — the same allowance we make for LBGAT. On CIFAR-10 the port reproduces HAT’s published no-extra-data figure to within 0.31 clean and 0.27 AutoAttack (84.90/49.08 reported, ResNet-18), which is what licenses reporting its CIFAR-100 row at all: HAT publishes no CIFAR-100 result without additional data, its 61.50/28.88 being trained with DDPM-generated samples, so that column has no published figure to be read against.

ADR is run on separate hardware and its row is filled from there. Two things have to hold when it is. ADR reports two configurations — AT+ADR at 56.10/26.87 on CIFAR-100 and AT+WA+AWP+ADR at 57.36/28.50 — and the row has to name which, the second being the one component-matched to our recipe, which carries both weight averaging and AWP. ADR is also self-distillation: its teacher is an exponential moving average of the student, no external network is read, and the student trains from scratch, so its row inherits nothing of our natural teacher.

What the published tables are blocked by. Table 27 and Table 28 separate their rows by network, and the separations are not cosmetic. RPAT’s “ResNet-18” is ResNet(Bottleneck, [2,2,2,2]) in its released code, 14.35M parameters against the 11.27M of the standard BasicBlock network used by us and by ADR; they also ship a `pre_resnet18` at 11.27M and use PreActResNet-18 explicitly elsewhere, so the distinction is theirs. HAT’s Tiny-ImageNet row is PreActResNet-18: ADR lists it under ResNet-18, but the figures are quoted from HAT’s own Table 9, whose caption says PreAct, and HAT’s code refuses anything else on that dataset — same parameter count as ours, different network, and its two companions from that table are shown beside it. F²AT’s plain SAT baseline reaches 31.52/9.37 where ADR’s plain AT reaches 45.87/18.06, a gap no difference of method explains, so its rows are separated too. TE’s architecture we could not verify — it is ADR’s attribution to a paper we do not have — and it is marked accordingly. ADR trains Tiny-ImageNet for 80 epochs against our 100, which favours us and is stated here rather than left implicit. LBGAT reports that dataset without AutoAttack and B-MTARD reports it on MobileNet-v2, so neither appears; CURE, ARREST and Generalist++ do not report it at all.

IGDM is reported at $\alpha = 1$. Its published $\alpha = 20$ diverges to chance accuracy on both datasets when the teacher is natural, as documented in Table 17.

Teacher-only KL with the training stack. This diagnostic combines probability-target distillation with WA and AWP; it is not a reproduction of a published method. Temperature is applied only to the teacher, and the best tested stacked setting, $\tau = 4$, gives 56.97/26.96 clean/AA; $\tau = 16$ gives 42.45/20.10. The comparison concerns this KL training design, not all logit-based supervision. In particular, fixed-head logit MSE is evaluated separately in Table 11, where it remains competitive with feature regression.

L PER-ATTACK METRICS

Table 26 gives every attack we measure for our own cells. The main tables report clean accuracy and AutoAttack because those are the two axes the trade-off is defined on, and because PGD-20 is not a faithful reading of this method.

Table 26: Per-attack accuracy (%) for our cells. ResNet-18, $\epsilon = 8/255$, final-epoch weight-averaged model, full test set.

Dataset	Cell	Clean	FGSM	PGD-20	CW	AA	NRR
CIFAR-100	CFA, head frozen (100 ep.)	62.17	36.91	32.37	30.93	28.86	39.42
	the same cell at seed 1	62.00	36.79	32.49	30.82	28.74	39.27
	ℓ_1 gradient norm	62.16	36.48	31.97	30.56	28.57	39.15
	CFA, head-KD term retained	62.35	38.80	36.26	30.65	28.68	39.29
	CFA, $\epsilon_{\text{tr}} = 6/255$	68.07	35.88	28.40	28.16	25.82	37.44
	CFA, $\epsilon_{\text{tr}} = 7/255$	65.61	36.32	30.34	29.64	27.44	38.70
	CFA, $\epsilon_{\text{tr}} = 8/255$	63.89	36.67	31.41	29.83	27.78	38.72
	CFA, $\epsilon_{\text{tr}} = 10/255$	59.99	36.50	32.82	30.53	28.77	38.89
	CFA, no WA / AWP	61.29	33.97	28.23	27.19	25.36	35.88
	Teacher-only KL, our stack	56.97	35.82	33.68	28.94	26.96	36.60
	RPAT++ (our reproduction)	55.93	34.46	32.44	29.37	27.36	36.74
CIFAR-10	CFA, $\epsilon_{\text{tr}} = 8.8/255$ (100 ep.)	84.96	59.58	52.98	53.94	51.74	64.31
	CFA, $\epsilon_{\text{tr}} = 6/255$	89.64	59.37	49.29	50.20	47.88	62.42
	CFA, $\epsilon_{\text{tr}} = 7/255$	88.30	59.59	51.43	52.64	50.27	64.07
	CFA, $\epsilon_{\text{tr}} = 8/255$	87.22	59.92	52.43	53.55	51.15	64.48
	CFA, $\epsilon_{\text{tr}} = 10/255$	83.29	58.98	53.25	53.88	51.79	63.87
	RPAT++ (our reproduction)	82.41	59.73	55.72	52.88	50.75	62.82

Why PGD-20 is not the metric here. The student inherits the teacher’s classifier and never retrains it. PGD-20 ascends the cross-entropy *through that classifier*, so it reads a head that was fitted for a different feature distribution; AutoAttack, which includes a targeted APGD, and CW_∞ , which attacks the logit margin directly, do not depend on that fit in the same way. The consequence is measurable on a controlled pair. The first two rows of Table 26 share a backbone objective, a schedule and an attack, and differ only in whether the classifier is refit:

	PGD-20	CW	AA
head-KD term retained	36.26	30.65	28.68
head frozen	32.63	30.66	28.77
difference	−3.63	+0.01	+0.09

Changing the classifier moves PGD-20 by 3.63 points and moves CW and AutoAttack by hundredths. The same pattern recurs across our ablations—the head-temperature sweep, the label-smoothing sweep, the directional-versus-raw comparison, and the radius sweep all show PGD-20 moving against CW and AutoAttack. We therefore report PGD-20 for completeness and read the trade-off on clean accuracy and AutoAttack, which is also what ADR and ARREST report in their main tables.

M PUBLISHED RESULTS, FOR CONTEXT

Table 2 contains only numbers we measured ourselves. Table 27 collects the published figures for the same and neighbouring methods, so that our runs can be placed against the literature without the two being mixed in one column. These are quoted as reported and were produced by other codebases, other schedules and other checkpoint-selection rules—RPAT selects its best checkpoint by PGD-20 and ADR by PGD-10, where every number of ours is the final epoch—so they should be compared within a source rather than across.

Table 27: Published results, ℓ_∞ , $\epsilon = 8/255$, quoted as reported. **The architecture changes between blocks** and is named in each heading; only the ResNet-18 blocks are comparable with Table 2. These were produced by other codebases, schedules and checkpoint-selection rules, so they should be compared within a source rather than across.

Method	Clean	AA	Avg	NRR	Source
<i>CIFAR-100, ResNet-18</i>					
PGD-AT	56.56	25.02	40.79	34.69	RPAT
TRADES	55.39	24.51	39.95	33.98	RPAT
MART	49.83	25.00	37.41	33.30	RPAT
Consistency-AT	58.53	25.39	41.96	35.42	RPAT
Consistency-AT + RPAT	60.33	26.31	43.32	36.64	RPAT
F ² AT	54.19	23.24	38.71	32.53	Qian et al. (2023)
Generalist++	62.97	23.96	43.47	34.71	Wang & Wang (2023)
ADR (AT + ADR)	56.10	26.87	41.48	36.34	ADR
ADR (AT + WA + AWP + ADR)	57.36	28.50	42.93	38.08	ADR
<i>CIFAR-10, ResNet-18</i>					
PGD-AT	82.78	44.63	63.71	57.99	CURE
TRADES	82.41	48.37	65.39	60.96	CURE
MART	80.70	47.49	64.10	59.79	CURE
ST-AT	83.10	50.50	66.80	62.82	CURE
IAD	80.63	50.17	65.40	61.85	CURE
Consistency-AT + RPAT	84.12	48.98	66.55	61.91	RPAT
Generalist++	89.09	46.07	67.58	60.73	Wang & Wang (2023)
ARREST	86.63	46.14	66.39	60.21	ARREST
CURE	86.76	49.69	68.23	63.19	CURE
ADR (AT + ADR)	82.41	50.38	66.40	62.53	ADR
<i>CIFAR-10, PreActResNet-18</i>					
WA (Izmailov et al., 2018)	83.50	49.89	66.70	62.46	RPAT
AWP (Wu et al., 2020)	81.11	50.09	65.60	61.93	RPAT
KD + SWA (Chen et al., 2021)	84.06	49.82	66.94	62.56	RPAT
TE (Dong et al., 2022)	82.04	50.12	66.08	62.23	RPAT
ReBAT (Wang et al., 2023)	82.09	50.72	66.41	62.70	RPAT
RPAT++ (Wang & Liu, 2025)	82.63	51.00	66.82	63.07	RPAT
<i>CIFAR-100, PreActResNet-18</i>					
WA (Izmailov et al., 2018)	57.26	25.83	41.55	35.60	RPAT
AWP (Wu et al., 2020)	54.10	25.16	39.63	34.35	RPAT
KD + SWA (Chen et al., 2021)	57.17	25.66	41.42	35.42	RPAT
TE (Dong et al., 2022)	56.41	25.84	41.13	35.44	RPAT
ReBAT (Wang et al., 2023)	56.13	27.60	41.87	37.00	RPAT
RPAT++ (Wang & Liu, 2025)	56.84	27.68	42.26	37.23	RPAT
<i>CIFAR-10, WideResNet-34-10</i>					
WA (Izmailov et al., 2018)	87.66	52.65	70.16	65.79	RPAT
MMA (Ding et al., 2020)	87.80	43.10	65.45	57.82	RPAT
AWP (Wu et al., 2020)	85.63	53.32	69.48	65.72	RPAT
KD + SWA (Chen et al., 2021)	87.45	53.59	70.52	66.46	RPAT
TE (Dong et al., 2022)	85.97	52.88	69.43	65.48	RPAT
ReBAT (Wang et al., 2023)	85.25	54.78	70.02	66.70	RPAT
ADR (Wu et al., 2024)	84.67	53.25	68.96	65.38	RPAT
CURE (Gowda et al., 2024)	87.05	52.10	69.58	65.19	RPAT
RPAT++ (Wang & Liu, 2025)	86.76	54.97	70.87	67.30	RPAT

Read against Table 2 and restricted to the ResNet-18 blocks, which are the ones our numbers are comparable with, our CFA row exceeds every published Avg and NRR on both datasets: 64.82/28.04 at Avg 46.43 on CIFAR-100 and 87.37/51.33 at Avg 69.35 on CIFAR-10.

The restriction is not a convenience. The WideResNet-34-10 block exceeds ours on both criteria—RPAT++ reaches Avg 70.87 and NRR 67.30 there—because a 48.3M network is a different model, not a better method, and Tables 3 and 19 are where that comparison belongs. The same caution applies to the PreActResNet-18 blocks in the other direction: RPAT++ falls to 82.63/51.00 there, and reading that as a defeat for RPAT would be the identical error. We state all of this as a comparison across codebases and architectures rather than as a controlled result, which is what Table 2 is for.

Table 28: Published Tiny-ImageNet-200 results, ResNet-18, ℓ_∞ , $\epsilon = 8/255$, quoted as reported. Our row is above all of them on both axes. The blocks are separated because the networks are not the same, and Appendix K says how each differs.

Method	Clean	AA	Avg	NRR	Source
CFA (ours)	55.16	20.54	37.85	29.93	—
AT + WA + ADR	48.55	20.23	34.39	28.56	ADR
AT + WA + AWP + ADR	48.27	20.12	34.20	28.40	ADR
TRADES + WA + AWP + ADR	51.38	19.48	35.43	28.25	ADR
TRADES + WA + ADR	51.99	19.17	35.58	28.01	ADR
AT + WA + AWP	48.61	19.58	34.09	27.92	ADR
AT + WA	49.10	19.30	34.20	27.71	ADR
TE*	47.46	18.29	32.88	26.40	ADR
AT	45.87	18.06	31.96	25.92	ADR
TRADES	48.49	17.35	32.92	25.56	ADR
<i>PreActResNet-18 — same parameter count, different network</i>					
HAT	52.60	18.14	35.37	26.98	HAT
TRADES	48.25	17.17	32.71	25.35	HAT
AT	47.76	17.92	32.84	26.02	HAT
<i>Bottleneck ResNet-18 (14.35M) and an unstated setup — see caption</i>					
Consistency-AT + RPAT	49.74	18.84	34.29	27.33	RPAT
PGD-AT + RPAT	47.68	17.77	32.73	25.89	RPAT
Consistency-AT	46.54	17.60	32.07	25.54	RPAT
MART	39.70	17.18	28.44	23.98	RPAT
F ² AT	40.54	13.13	26.84	19.84	F ² AT
SEAT	35.51	11.37	23.44	17.22	F ² AT
SAT	31.52	9.37	20.45	14.45	F ² AT

N THE CURE REPRODUCTION

Seven runs of the official CURE repository, from its own code and our shared CIFAR data, none of which reproduce the published 86.76/49.69 on CIFAR-10. They fall short on *different* axes: the closest on standard accuracy reaches 86.11 at AA 40.65, the closest on robustness AA 45.63 at clean 81.15, while the published pair has both. Running the repository unmodified gives 65.03/29.74. Computing its RGP prominence mask on absolute values raises AA by 15.9 points (29.74 $\rightarrow$ 45.63), which localizes the discrepancy to that step without closing it.

We re-evaluated all five surviving checkpoints with our own pipeline and it agrees with the repository’s own evaluation to within 0.05 AutoAttack on every one, so the gap is in training rather than in measurement. For contrast, RPAT++ reproduces to within 0.31 AutoAttack of its published numbers on both datasets under the same conditions. The CURE row of Table 27 is therefore the paper’s own number, and CURE is omitted from Table 2.

O A CORRECTION TO THE AWP PROXY, AND WHAT IT MOVED

AWP ascends an auxiliary objective in weight space before each descent step. In our implementation the closure used for that ascent had drifted from the closure used for descent, in two ways. It

retained a head-distillation term that `freeze_head` removes from training, so AWP was maximizing a quantity partly composed of a term the model never optimizes; and a detach intended for the normalized feature was applied on a path that feeds the unnormalized one. Measured on CIFAR-100 before the fix, the spurious term was 7.15 against the anchor’s 4.15 and contributed 40.8% of the AWP backbone gradient, at cosine 0.918 to the correct direction. Separately, the head was frozen *after* the proxy was copied, which is harmless only because the proxy’s first step is delayed by the AWP warmup.

We report this because it is the kind of defect that changes a paper’s numbers, and because here it did not. Re-running the reported recipe with both closures agreeing gives 62.17/28.86 against 62.65/28.77 on CIFAR-100 and 84.96/51.74 against 85.58/51.79 on CIFAR-10: NRR moves by 0.01 and 0.22, no ordering in any table changes, and clean accuracy moves more than robustness does. AWP is evidently insensitive to a large, well-aligned perturbation of its ascent direction, which is consistent with its role here as a regularizer of sharpness rather than a precisely targeted objective. All numbers reported for the reported recipe are post-correction. Two exceptions are stated where they appear: the head-freeze comparison in Appendix L and the ϵ_{train} sweep in Table 15 were measured before it. In each, every arm shares the defect, so the comparison is internally consistent even though its absolute values differ from the corrected ones by a few hundredths.

REFERENCES

- Anish Athalye, Nicholas Carlini, and David Wagner. Obfuscated gradients give a false sense of security: Circumventing defenses to adversarial examples. In *International conference on machine learning*, pp. 274–283. PMLR, 2018.
- Yogesh Balaji, Tom Goldstein, and Judy Hoffman. Instance adaptive adversarial training: Improved accuracy tradeoffs in neural nets. *arXiv preprint arXiv:1910.08051*, 2019.
- Tianlong Chen, Zhenyu Zhang, Sijia Liu, Shiyu Chang, and Zhangyang Wang. Robust overfitting may be mitigated by properly learned smoothening. In *International Conference on Learning Representations*, 2021.
- Minhao Cheng, Qi Lei, Pin-Yu Chen, Inderjit Dhillon, and Cho-Jui Hsieh. Customized adversarial training for improved robustness. In *IJCAI*, 2022.
- Jiequan Cui, Shu Liu, Liwei Wang, and Jiaya Jia. Learnable boundary guided adversarial training. In *ICCV*, 2021.
- Gavin Weiguang Ding, Yash Sharma, Kry Yik Chau Lui, and Ruitong Huang. MMA training: Direct input space margin maximization through adversarial training. In *ICLR*, 2020.
- Yinpeng Dong, Ke Xu, Xiao Yang, Tianyu Pang, Zhijie Deng, Hang Su, and Jun Zhu. Exploring memorization in adversarial training. In *International Conference on Learning Representations*, 2022.
- Micah Goldblum, Liam Fowl, Soheil Feizi, and Tom Goldstein. Adversarially robust distillation. In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 34, pp. 3996–4003, 2020.
- Shruthi Gowda, Bahram Zonooz, and Elahe Arani. Conserve-update-revise to cure generalization and robustness trade-off in adversarial training. In *ICLR*, 2024.
- Bo Huang, Mingyang Chen, Yi Wang, Junda Lu, Minhao Cheng, and Wei Wang. Boosting accuracy and robustness of student models via adaptive adversarial distillation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 24668–24677, 2023.
- Pavel Izmailov, Dmitrii Podoprikin, Timur Garipov, Dmitry Vetrov, and Andrew Gordon Wilson. Averaging weights leads to wider optima and better generalization. In *Conference on Uncertainty in Artificial Intelligence*, 2018.
- Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009.

1458 Ya Le and Xuan Yang. Tiny imagenet visual recognition challenge. CS231n course project report,
1459 Stanford University, 2015.

1460

1461 Hongsin Lee, Seungju Cho, and Changick Kim. Indirect gradient matching for adversarial robust
1462 distillation. In *ICLR*, 2025.

1463 Aleksander Madry, Aleksandar Makelov, Ludwig Schmidt, Dimitris Tsipras, and Adrian Vladu.
1464 Towards deep learning models resistant to adversarial attacks. *arXiv preprint arXiv:1706.06083*,
1465 2017.

1466

1467 Tianyu Pang, Xiao Yang, Yinpeng Dong, Hang Su, and Jun Zhu. Bag of tricks for adversarial
1468 training. *arXiv preprint arXiv:2010.00467*, 2020.

1469 Yaguan Qian, Chenyu Zhao, Zhaoquan Gu, Bin Wang, Shouling Ji, Wei Wang, Boyang Zhou, and
1470 Pan Zhou. F²AT: Feature-focusing adversarial training via disentanglement of natural and per-
1471 turbed patterns. *arXiv preprint arXiv:2310.14561*, 2023.

1472

1473 Rahul Rade and Seyed-Mohsen Moosavi-Dezfooli. Reducing excessive margin to achieve a bet-
1474 ter accuracy vs. robustness trade-off. In *International Conference on Learning Representations*,
1475 2022.

1476

1477 Leslie Rice, Eric Wong, and J. Zico Kolter. Overfitting in adversarially robust deep learning. In
1478 *ICML*, 2020.

1479 Satoshi Suzuki, Shin’ya Yamaguchi, Shoichiro Takeda, Sekitoshi Kanai, Naoki Makishima, Atsushi
1480 Ando, and Ryo Masumura. ARREST: Adversarial finetuning with latent representation constraint
1481 to mitigate the accuracy-robustness tradeoff. In *ICCV*, 2023.

1482

1483 Jihoon Tack, Sihyun Yu, Jongheon Jeong, Minseon Kim, Sung Ju Hwang, and Jinwoo Shin. Con-
1484 sistency regularization for adversarial robustness. In *Proceedings of the AAAI Conference on*
1485 *Artificial Intelligence*, volume 36, pp. 8414–8422, 2022.

1486

1487 Dimitris Tsipras, Shibani Santurkar, Logan Engstrom, Alexander Turner, and Aleksander Madry.
1488 Robustness may be at odds with accuracy. *arXiv preprint arXiv:1805.12152*, 2018.

1489

1490 Hongjun Wang and Yisen Wang. Generalist: Decoupling natural and robust generalization. In *Pro-*
1491 *ceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 20554–
1492 20563, 2023.

1493

1494 Yanyun Wang and Li Liu. Failure cases are better learned but boundary says sorry: Facilitating
1495 smooth perception change for accuracy-robustness trade-off in adversarial training. In *ICCV*,
1496 2025.

1497

1498 Yifei Wang, Liangchen Li, Jiansheng Yang, Zhouchen Lin, and Yisen Wang. Balance, imbalance,
1499 and rebalance: Understanding robust overfitting from a minimax game perspective. In *NeurIPS*,
1500 2023.

1501

1502 Yisen Wang, Difan Zou, Jinfeng Yi, James Bailey, Xingjun Ma, and Quanquan Gu. Improving
1503 adversarial robustness requires revisiting misclassified examples. In *International Conference on*
1504 *Learning Representations*, 2020.

1505

1506 Dongxian Wu, Shu-Tao Xia, and Yisen Wang. Adversarial weight perturbation helps robust gener-
1507 alization. *Advances in Neural Information Processing Systems*, 33:2958–2969, 2020.

1508

1509 Yu-Yu Wu, Hung-Jui Wang, and Shang-Tse Chen. Annealing self-distillation rectification improves
1510 adversarial training. In *ICLR*, 2024.

1511

Hongyang Zhang, Yaodong Yu, Jiantao Jiao, Eric Xing, Laurent El Ghaoui, and Michael Jordan.
Theoretically principled trade-off between robustness and accuracy. In *International conference*
on machine learning, pp. 7472–7482. PMLR, 2019.

Shiji Zhao, Xizhe Wang, and Xingxing Wei. Mitigating accuracy-robustness trade-off via balanced
multi-teacher adversarial distillation. *IEEE TPAMI*, 2024.

1512 Jianing Zhu, Jiangchao Yao, Bo Han, Jingfeng Zhang, Tongliang Liu, Gang Niu, Jingren Zhou,
1513 Jianliang Xu, and Hongxia Yang. Reliable adversarial distillation with unreliable teachers. *arXiv*
1514 *preprint arXiv:2106.04928*, 2021.

1515 Bojia Zi, Shihao Zhao, Xingjun Ma, and Yu-Gang Jiang. Revisiting adversarial robustness distil-
1516 lation: Robust soft labels make student better. In *Proceedings of the IEEE/CVF International*
1517 *Conference on Computer Vision*, pp. 16443–16452, 2021.

1518

1519

1520

1521

1522

1523

1524

1525

1526

1527

1528

1529

1530

1531

1532

1533

1534

1535

1536

1537

1538

1539

1540

1541

1542

1543

1544

1545

1546

1547

1548

1549

1550

1551

1552

1553

1554

1555

1556

1557

1558

1559

1560

1561

1562

1563

1564

1565